

# DBMS – PRODUCT AND CATEGORY MANAGEMENT

## HUFT – E-Commerce Order Management Database System

### 1.DATABASE DESIGN:

#### 1.1 CATEGORY TABLE :

| Attribute     | Description        |
|---------------|--------------------|
| Category_ID   | Unique category ID |
| Category_Name | Category name      |

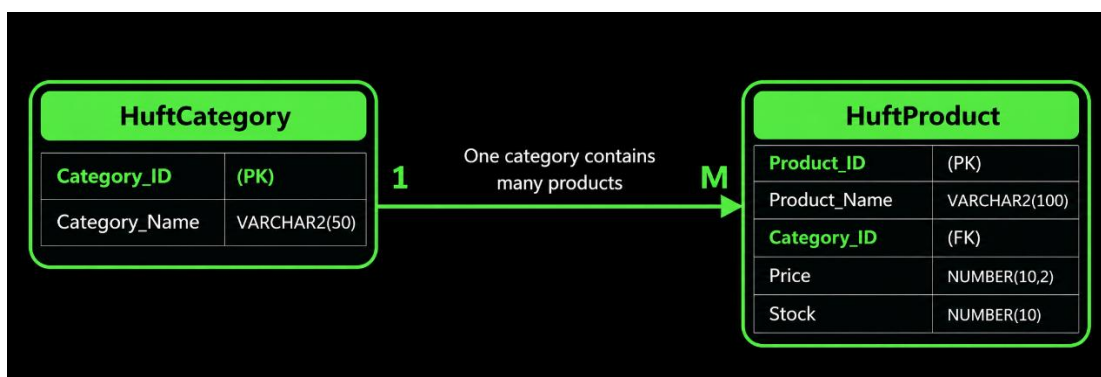
#### 1.2 PRODUCT TABLE

| Attribute    | Description       |
|--------------|-------------------|
| Product_ID   | Unique product ID |
| Product_Name | Product name      |
| Price        | Product price     |
| Stock        | Available stock   |
| Category_ID  | Category ID       |

### 2.\_PRIMARY AND FOREIGN KEY RELATIONSHIP

HuftCategory → HuftProduct

- HuftCategory.Category\_ID → Primary Key
- HuftProduct.Product\_ID → Primary Key
- HuftProduct.Category\_ID → Foreign Key referencing HuftCategory.Category\_ID
- Relationship: One-to-Many (1:M)



### 3.TABLE CREATION

### 3.1 CREATE CATEGORY TABLE

```
SQL> CREATE TABLE HuftCategory (  
  2 Category_ID NUMBER PRIMARY KEY,  
  3 Category_Name VARCHAR2(50)  
  4 );
```

Table created.

### 3.2 CREATE PRODUCT TABLE

```
PS D:\Users\Nandhini\Desktop\product> sqlplus  
SQL> CREATE TABLE HuftProduct (  
  2 Product_ID NUMBER PRIMARY KEY,  
  3 Product_Name VARCHAR2(100),  
  4 Category_ID NUMBER,  
  5 Price NUMBER(10,2),  
  6 Stock NUMBER,  
  7 CONSTRAINT fk1_product_category  
  8 FOREIGN KEY (Category_ID)  
  9 REFERENCES Category(Category_ID)  
 10 );
```

Table created.

## 4.DATA INSERTION

### 4.1 CATEGORY RECORDS

```
SQL> INSERT INTO HuftCategory VALUES (1, 'Dog Food');  
  
1 row created.  
  
SQL> INSERT INTO HuftCategory VALUES (2, 'Cat Food');  
  
1 row created.  
  
SQL> INSERT INTO HuftCategory VALUES (3, 'Accessories');  
  
1 row created.  
  
SQL> INSERT INTO HuftCategory VALUES (4, 'Grooming');  
  
1 row created.  
  
SQL> INSERT INTO HuftCategory VALUES (5, 'Toys');  
  
1 row created.
```

### 4.2 PRODUCT RECORDS

```

SQL> INSERT INTO HuftProduct VALUES (106, 'Cat Litter', 2, 550.00, 30);

1 row created.

SQL> INSERT INTO HuftProduct VALUES (107, 'Dental Chew Sticks', 1, 399.00, 45);

1 row created.

SQL> INSERT INTO HuftProduct VALUES (108, 'Pet Feeding Bowl', 3, 275.00, 20);

1 row created.

SQL> INSERT INTO HuftProduct VALUES (109, 'Puppy Training Pads', 1, 499.00, 55);

1 row created.

SQL> INSERT INTO HuftProduct VALUES (110, 'Pet Conditioner', 4, 425.00, 30);

```

## 5.DATA RETRIVAL

### 5.1 DISPLAY CATEGORY RECORDS

```
SQL> SELECT * FROM HuftCategory;
```

| CATEGORY_ID | CATEGORY_NAME |
|-------------|---------------|
| 1           | Dog Food      |
| 2           | Cat Food      |
| 3           | Accessories   |
| 4           | Grooming      |
| 5           | Toys          |

### 5.2 DISPLAY PRODUCT RECORDS

```
SQL> SELECT *FROM HuftProduct;
```

| PRODUCT_ID | PRODUCT_NAME        | CATEGORY_ID | PRICE | STOCK |
|------------|---------------------|-------------|-------|-------|
| 101        | Chicken Dog Food    | 1           | 599   | 50    |
| 102        | Grain Free Cat Food | 2           | 749   | 35    |
| 103        | Leather Dog Collar  | 3           | 450   | 25    |

104  
Pet Grooming Shampoo

PRODUCT\_ID

PRODUCT\_NAME

| CATEGORY_ID | PRICE | STOCK |
|-------------|-------|-------|
| 4           | 350   | 40    |

106  
Cat Litter

|   |     |    |
|---|-----|----|
| 2 | 550 | 30 |
|---|-----|----|

107  
PRODUCT\_ID

PRODUCT\_NAME

| CATEGORY_ID        | PRICE | STOCK |    |
|--------------------|-------|-------|----|
| Dental Chew Sticks | 1     | 399   | 45 |

108  
Pet Feeding Bowl

|   |     |    |
|---|-----|----|
| 3 | 275 | 20 |
|---|-----|----|

PRODUCT\_ID

PRODUCT\_NAME

| CATEGORY_ID | PRICE | STOCK |
|-------------|-------|-------|
|-------------|-------|-------|

|                     |   |     |    |
|---------------------|---|-----|----|
| 109                 |   |     |    |
| Puppy Training Pads | 1 | 499 | 55 |

110  
Pet Conditioner

|   |     |    |
|---|-----|----|
| 4 | 425 | 30 |
|---|-----|----|

PRODUCT\_ID

PRODUCT\_NAME

| CATEGORY_ID | PRICE | STOCK |
|-------------|-------|-------|
|-------------|-------|-------|

112  
Cat Grooming Brush

|   |     |    |
|---|-----|----|
| 4 | 325 | 28 |
|---|-----|----|

113  
Dog Harness

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    PRICE    STOCK
-----
          3      699      22

          114
Premium Dog Treats
          1      450      40

```

12 rows selected.

## 6.PRODUCT UPDATE

```

SQL> UPDATE HuftProduct
  2  SET Price = 650
  3  WHERE Product_ID = 101;

1 row updated.

```

```

SQL> UPDATE HuftProduct
  2  SET Stock = 80
  3  WHERE Product_ID = 102;

1 row updated.

```

## 7.PRODUCT DELETION

```

SQL> DELETE FROM HuftProduct
  2  WHERE Category_ID = 4;

3 rows deleted.

```

```

SQL> DELETE FROM HuftCategory
  2  WHERE Category_ID = 5;

1 row deleted.

```

## 8.CATEGORY UPDATE

```

SQL> UPDATE HuftCategory
  2  SET Category_Name = 'Premium Dog Food'
  3  WHERE Category_ID = 1;

1 row updated.

```

```

SQL> UPDATE HuftCategory
  2  SET Category_Name = 'Pet Accessories'
  3  WHERE Category_ID = 3;

1 row updated.

```

## 9. CATEGORY-WISE PRODUCT REPORT

```
SQL> SELECT
  2     c.Category_ID,
  3     c.Category_Name,
  4     p.Product_ID,
  5     p.Product_Name,
  6     p.Price,
  7     p.Stock
  8 FROM HuftCategory c
  9 JOIN HuftProduct p
 10 ON c.Category_ID = p.Category_ID
 11 ORDER BY c.Category_Name, p.Product_Name;
```

| CATEGORY_ID | CATEGORY_NAME | PRODUCT_ID |
|-------------|---------------|------------|
| 2           | Cat Food      | 106        |
|             | Cat Litter    |            |
| 550         | 30            |            |

|     |                     |     |
|-----|---------------------|-----|
| 2   | Cat Food            | 102 |
|     | Grain Free Cat Food |     |
| 749 | 80                  |     |

| CATEGORY_ID | CATEGORY_NAME   | PRODUCT_ID |
|-------------|-----------------|------------|
| 3           | Pet Accessories | 113        |
|             | Dog Harness     |            |
| 699         | 22              |            |

|   |                    |     |
|---|--------------------|-----|
| 3 | Pet Accessories    | 103 |
|   | Leather Dog Collar |     |

| CATEGORY_ID | CATEGORY_NAME    | PRODUCT_ID |
|-------------|------------------|------------|
| 450         | 25               |            |
| 3           | Pet Accessories  | 108        |
|             | Pet Feeding Bowl |            |
| 275         | 20               |            |
| 1           | Premium Dog Food | 101        |

| CATEGORY_ID | CATEGORY_NAME | PRODUCT_ID |
|-------------|---------------|------------|
|-------------|---------------|------------|

| PRODUCT_NAME |
|--------------|
|--------------|

| PRICE | STOCK |
|-------|-------|
|-------|-------|

|                  |  |
|------------------|--|
| Chicken Dog Food |  |
|------------------|--|

|     |    |
|-----|----|
| 650 | 50 |
|-----|----|

|                    |  |
|--------------------|--|
| 1 Premium Dog Food |  |
|--------------------|--|

|     |
|-----|
| 107 |
|-----|

|                    |  |
|--------------------|--|
| Dental Chew Sticks |  |
|--------------------|--|

|     |    |
|-----|----|
| 399 | 45 |
|-----|----|

| CATEGORY_ID | CATEGORY_NAME |
|-------------|---------------|
|-------------|---------------|

| PRODUCT_ID |
|------------|
|------------|

| PRODUCT_NAME |
|--------------|
|--------------|

| PRICE | STOCK |
|-------|-------|
|-------|-------|

|                    |  |
|--------------------|--|
| 1 Premium Dog Food |  |
|--------------------|--|

|     |
|-----|
| 114 |
|-----|

|                    |  |
|--------------------|--|
| Premium Dog Treats |  |
|--------------------|--|

|     |    |
|-----|----|
| 450 | 40 |
|-----|----|

|                    |  |
|--------------------|--|
| 1 Premium Dog Food |  |
|--------------------|--|

|     |
|-----|
| 109 |
|-----|

|                     |  |
|---------------------|--|
| Puppy Training Pads |  |
|---------------------|--|

|     |    |
|-----|----|
| 499 | 55 |
|-----|----|

| CATEGORY_ID | CATEGORY_NAME | PRODUCT_ID |
|-------------|---------------|------------|
|-------------|---------------|------------|

| PRODUCT_NAME |
|--------------|
|--------------|

| PRICE | STOCK |
|-------|-------|
|-------|-------|

9 rows selected.

## 10.CATEGORY-WISE TOTAL PRODUCTS REPORT

```
SQL> SELECT
  2     c.Category_ID,
  3     c.Category_Name,
  4     COUNT(p.Product_ID) AS Total_Products
  5 FROM HuftCategory c
  6 LEFT JOIN HuftProduct p
  7 ON c.Category_ID = p.Category_ID
  8 GROUP BY c.Category_ID, c.Category_Name
  9 ORDER BY c.Category_ID;
```

| CATEGORY_ID | CATEGORY_NAME | TOTAL_PRODUCTS |
|-------------|---------------|----------------|
|-------------|---------------|----------------|

|   |          |   |
|---|----------|---|
| 1 | Dog Food | 4 |
|---|----------|---|

|   |          |   |
|---|----------|---|
| 2 | Cat Food | 2 |
|---|----------|---|

|   |             |   |
|---|-------------|---|
| 3 | Accessories | 3 |
|---|-------------|---|

|   |          |   |
|---|----------|---|
| 4 | Grooming | 3 |
|---|----------|---|

|   |      |   |
|---|------|---|
| 5 | Toys | 0 |
|---|------|---|